

An Extended Complex Kalman Filter for Frequency Measurement of Distorted Signals

P. K. Dash, R. K. Jena, G. Panda, and Aurobinda Routray

Abstract—The design of an extended complex Kalman filter for the measurement of power system frequency has been presented in this paper. The design principles and the validity of the model have been outlined. A complex model has been developed to track a distorted signal that belongs to a power system. The model inherently takes care of the frequency measurement along with the amplitude and phase of the signals. The theory has been applied to standard test signals representing the worst-case measurement and network conditions in a typical power system. The proposed algorithm is suitable for real-time applications where the measurement noise and other disturbances are high. The complex quantities can be conveniently handled using a floating point processor. Comparison of the results of the proposed method with those obtained from a real extended Kalman filter reveals the superior performance of the former method.

Index Terms—Covariance setting, extended complex Kalman filter, frequency estimation, noise rejection, power system measurements.

I. INTRODUCTION

IN THIS paper, the method presented deals with the measurement of the parameters of a power system signal which is usually contaminated with noise and high disturbances. The amplitude and phase estimation of a digitized signal has been an important area of research for the past several years, and the methods have been almost standardized for signals with known frequencies. However, if the frequency is not known *a priori* it becomes a formidable task to accurately measure the amplitude and phase. It has been a perpetual problem even now to correctly estimate the frequency of an incoming signal from its sampled values under high noise conditions. The reason for this is the association of severe nonlinearity in the modeling process. On the other hand the philosophy of protection for power systems has been undergoing a change. The modern protective relays employ sophisticated signal processing algorithms to accurately estimate the system conditions from the measured voltage and current signals. The pitfalls in these schemes stem from the frequency measurement algorithm which, under transient and abnormal conditions, cannot sense the correct value.

Many algorithms have been reported in the literature for measurement and estimation of frequency.[1]–[4]. A comparative study among four different trackers has been outlined in [5]. In this paper, the performances of 1) an adaptive notch filter, 2) a multiple frequency tracker, 3) an adaptive IIR filter, and 4)

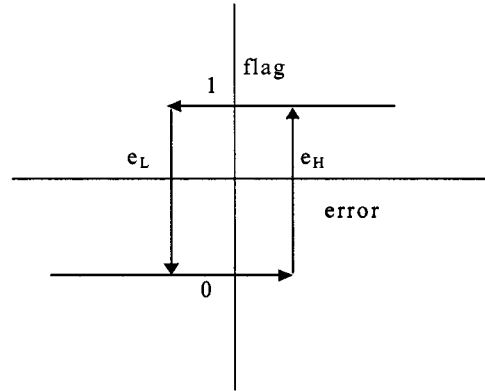


Fig. 1. The hysteresis band for resetting the covariance matrix.

a hyperstable adaptive line enhancer have been presented. The complex Kalman filtering has been used in [6] to estimate the frequency of the signals corrupted with white noise. The present paper is based on the fundamental work carried out in [6]. However, in practice the real and imaginary signals can not be obtained simultaneously. Therefore, keeping in view the measurement of power system signals, the signal model has been modified. The output equation is rewritten to calculate the real signal.

II. SIGNAL MODEL

The nonlinear state space description of the power system signal can have various forms. Numerous linear as well as nonlinear models have been proposed to estimate the amplitude, phase and frequency of a single sinusoid. This signal can also be represented by a complex model. With availability of floating point DSP processors the computation in the complex domain is no longer a difficult task. Moreover, the complex representation is much simpler and direct as far as the frequency measurement is concerned. Once the signal model is established the extended Kalman filter theory [7] can be applied to identify the filter equations and computational steps.

Let an observation signal y_k at time t_k be a sum z_k of M sinusoids with additive noise v_k

$$y_k = z_k + v_k \quad k = 1, 2, 3, \dots, N \quad (1)$$

where

$$z_k = \sum_{i=1}^M a_i \sin(\omega_i t_k + \phi_i) \quad \omega_i = 2\pi f_i, t_k = k\Delta t \quad (2)$$

in which a_i , f_i , and ϕ_i are amplitude, frequency and the phase of the i th sinusoid, respectively. The observation noise v_k is a Gaussian white noise with zero-mean and variance σ_v^2 .

Manuscript received June 15, 1998; revised March 4, 2000.

P. K. Dash, R. K. Jena, and G. Panda are with the Regional Engineering College, Rourkela - 769008, India.

A. Routray is with the Indian Institute of Technology, Kharagpur, India.

Publisher Item Identifier S 0018-9456(00)04805-1.

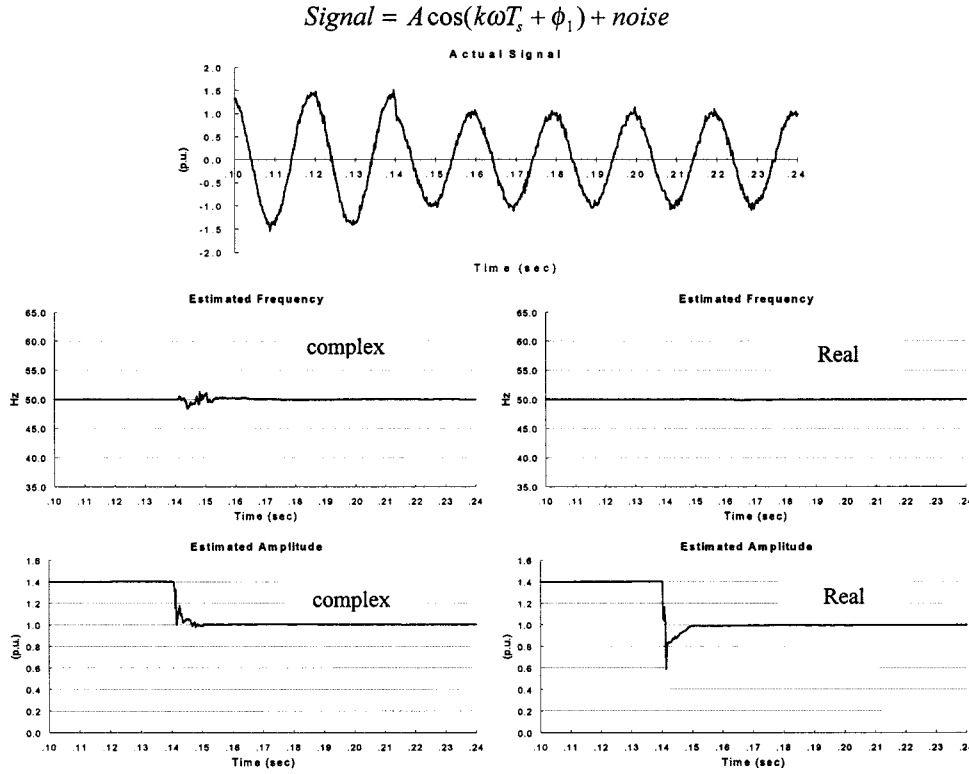


Fig. 2. Step down in amplitude for signal 1.

In the case of power system signals, the percentage of frequency components other than fundamental is low. Therefore these harmonic components need not be considered in the model.

For such systems, (2) reduces to a single sinusoid

$$z_k = a_1 \sin(k\omega_1 T_s + \phi_1) \quad (3)$$

where T_s = sampling time, and ω_1 = the fundamental angular frequency

The observation signal can be represented in an autoregressive complex form as follows:

$$\begin{bmatrix} \alpha \\ u_{k+1} \\ u_{k+1}^* \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \alpha & 0 \\ 0 & 0 & \frac{1}{\alpha} \end{bmatrix} \begin{bmatrix} \alpha \\ u_k \\ u_k^* \end{bmatrix} \quad (4)$$

$$y_k = [0 \quad 0.5 \quad -0.5] \begin{bmatrix} \alpha \\ u_k \\ u_k^* \end{bmatrix} + v_k \quad (5)$$

where

$$\alpha = \exp(j\omega_1 T_s) \quad (6)$$

$$u_k = a_1 \exp(jk\omega_1 T_s + j\phi_1) \quad (7)$$

$$u_k^* = a_1 \exp(-jk\omega_1 T_s - j\phi_1) \quad (8)$$

v_k = measurement noise.

The above nonlinear process can be represented as

$$x_{k+1} = f(x_k) \quad (9)$$

$$y_{k+1} = Hx_k + v_k \quad (10)$$

where

$$x(k) = [\alpha \quad u_k \quad u_k^*]^T \quad (11)$$

$$f(x_k) = \begin{bmatrix} \alpha & \alpha u_k & \frac{u_k^*}{\alpha} \end{bmatrix}^T \quad (12)$$

$$H = [0 \quad 0.5 \quad -0.5]. \quad (13)$$

Linearizing the above system and applying the extended complex Kalman filter to the first-order system, a nonlinear recursive filter for estimating a single complex sinusoid and its frequency in white noise is obtained as follows [6]:

$$\hat{x}_{k|k} = \hat{x}_{k|k-1} + K_k(y_k - H\hat{x}_{k|k-1}) \quad (14)$$

$$\hat{x}_{k+1|k} = f(\hat{x}_{k|k}) \quad (15)$$

$$K_k = \hat{P}_{k|k-1} H^{*T} [H \hat{P}_{k|k-1} H^{*T} + 1]^{-1} \quad (16)$$

$$\hat{P}_{k|k} = \hat{P}_{k|k-1} - K_k H \hat{P}_{k|k-1} \quad (17)$$

$$\hat{P}_{k+1|k} = F_k \hat{P}_{k|k} F_k^{*T} \quad (18)$$

where

$$F_k = \frac{\partial f(x_k)}{\partial x_k} \bigg|_{x_k = \hat{x}_{k|k}} = \begin{bmatrix} 1 & 0 & 0 \\ \hat{x}_{k|k}(2) & \hat{x}_{k|k}(1) & 0 \\ -\frac{\hat{x}_{k|k}(3)}{\hat{x}_{k|k}^2(1)} & 0 & \frac{1}{\hat{x}_{k|k}(1)} \end{bmatrix}$$

$$\hat{P}_{k|k} = \hat{\Sigma}_{k|k} / \sigma_v^2 \quad \hat{P}_{k+1|k} = \hat{\Sigma}_{k+1|k} / \sigma_v^2$$

$$\hat{\Sigma}_{k|k} = E\{(x_k - \hat{x}_{k|k})(x_k - \hat{x}_{k|k})^{*T}\}$$

$$\hat{\Sigma}_{k+1|k} = E\{(x_{k+1} - \hat{x}_{k+1|k})(x_{k+1} - \hat{x}_{k+1|k})^{*T}\}$$

$$\hat{x}_{1|0} = \bar{x}_1, \quad \hat{P}_{1|0} = \hat{\Sigma}_{1|0} / \sigma_v^2$$

$$\hat{\Sigma}_{1|0} = \Sigma_{x1}, \quad \Sigma_{x1} = E\{(x_1 - \bar{x}_1)(x_1 - \bar{x}_1)^{*T}\}$$

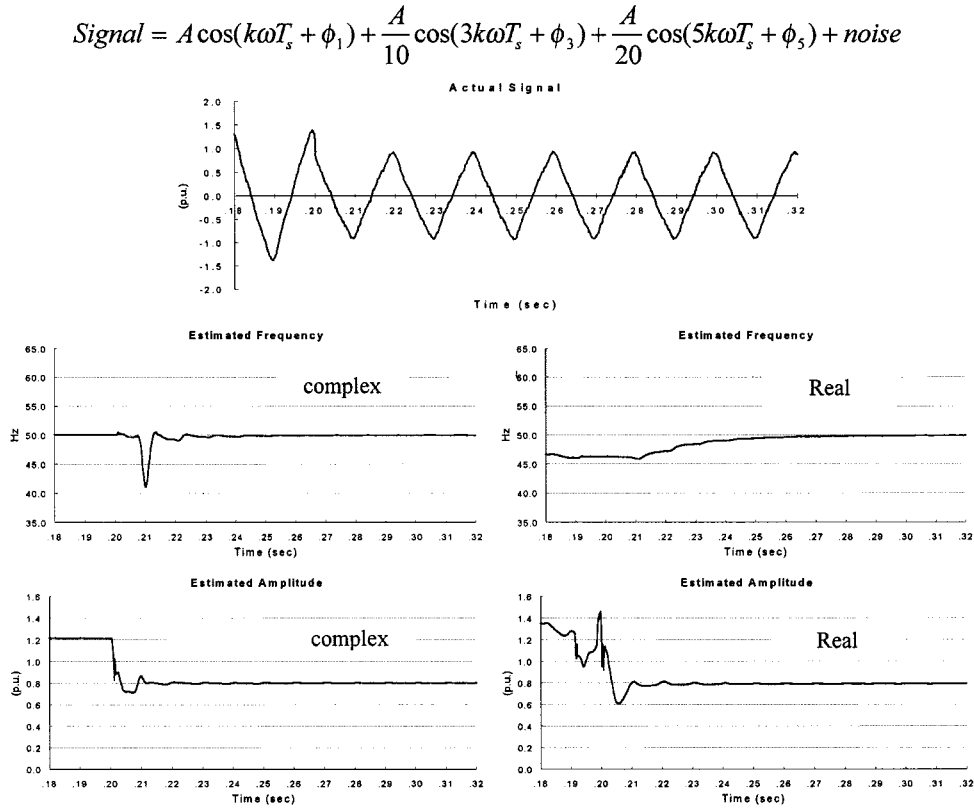


Fig. 3. Step up in amplitude for signal 2.

The filter is nonlinear and therefore the gain K_k and the covariance matrix $\hat{P}_{k|k}$ depend on the estimate $\hat{x}_{k|k}$ of the state vector x_k .

III. SIMULATION

The following tests have been performed to evaluate the performance of the above method.

Test Signal-I: Fundamental signal with white noise [signal-to-noise ratio (SNR) 13 dB]

$\text{Signal} = A \cos(k\omega T_s + \phi) + \text{Zero mean Gaussian white noise}$
of standard deviation 0.1

where $A = 1.0$

Test Signal-II: Fundamental signal with harmonics and white noise [SNR 37 dB]

$\text{Signal} = A \cos(k\omega T_s + \phi_1) + (A/10) \cos(3k\omega T_s + \phi_3)$
 $+ (A/20) \cos(5k\omega T_s + \phi_5)$
 $+ \text{noise as above with standard deviation 0.01}$

where $A = 1.0$

The problem with all Kalman filter algorithms is to reset the covariance matrix. After initial convergence the gain K_k and the covariance matrix P_{kk} settle to very small values. Subsequently when some of the parameters (amplitude, phase, and frequency) of the signal change, the covariance matrix has to be reset to quickly track these values. In this paper a method based on the magnitude of the error has been used.

The decision to set the covariance matrix to the initial value is based on a hysteresis type decision block. The hysteresis band

is determined by the amount of noise and the nature of convergence band. If the noise is estimated to be around 10% of the amplitude, then the hysteresis band is chosen to be 20–30% of the amplitude to avoid frequent resetting of the covariance matrix. A flag is set when the error exceeds the higher threshold e_H and is reset when the error falls below the lower threshold e_L . If the flag is 1 and any of the Kalman gains are very small, then the covariance is reset and the flag is also reset to 0 so that there is no immediate resetting of the covariance matrix (see Fig. 1).

Generally, the frequency variation in power systems is limited to ± 5 Hz. Therefore, for faster tracking the frequency is limited to 40 Hz on the lower side and 60 Hz on the upper side. This results in the stable operation of the filter and does not lead the filter wayward and lose track of the desired frequency under high noise conditions.

Two different types of test signals as above have been taken to study the effectiveness of the proposed algorithm. In each of the cases the fundamental amplitude and frequency are subjected to sudden changes, and under such conditions the tracking ability of the proposed filter is examined. The same tests are also carried out using an ordinary extended Kalman filter (see the Appendix). The comparative performance has been presented in Figs. 2–9.

Case-I: The amplitude is suddenly reduced from 1.4 p.u. to 1.0 p.u. The test is repeated for the above two types of signals, i.e., signal with and without harmonic components. For the latter the amplitude is reduced from 1.2 p.u. to 0.8 p.u. The signal without harmonics contains more noise as compared to the signal with harmonics. The instantaneous waveforms have been displayed in Figs. 2 and 3. The comparative study shows

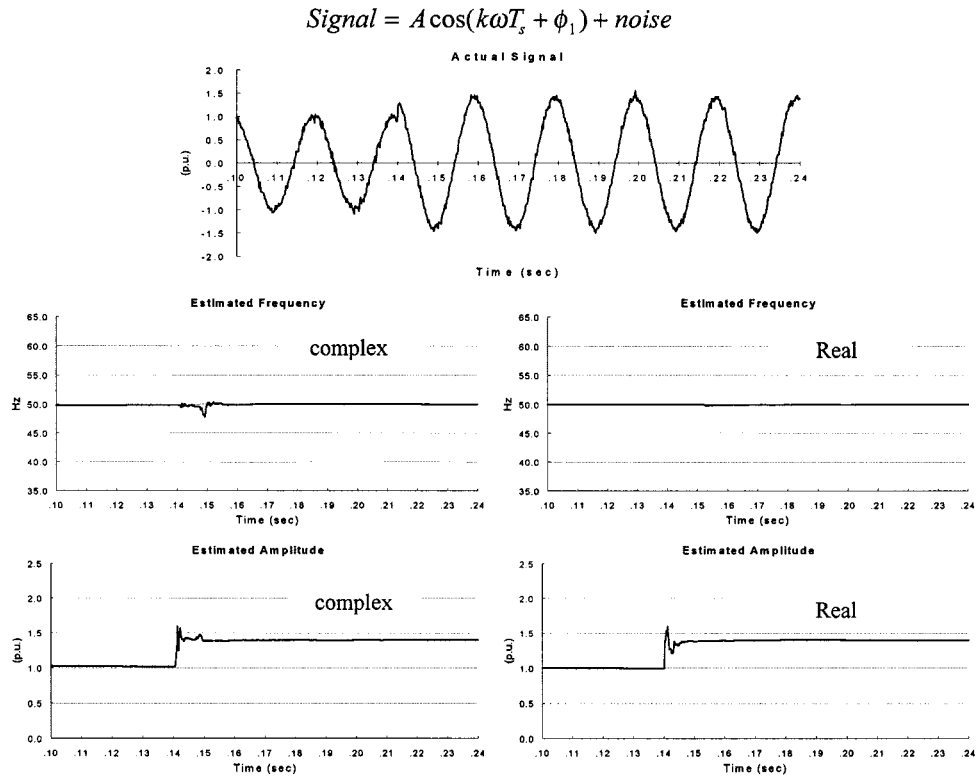


Fig. 4. Step up in amplitude for signal 1.

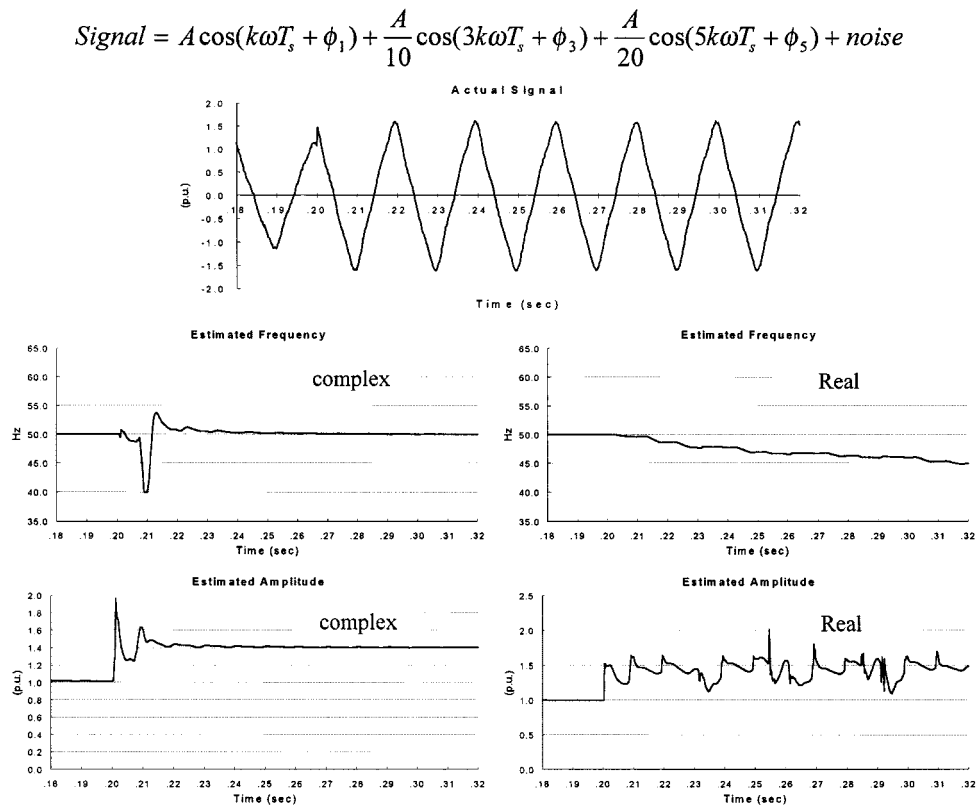


Fig. 5. Step up in amplitude for signal 2.

that the complex Kalman filter exhibits slightly inferior performance in estimating the amplitude and frequency in the case of

the signals without harmonics. Both the filters settle down at the new values within approximately half a cycle (about 0.01

$$\text{Signal} = A \cos(k\omega T_s + \phi_1) + \text{noise}$$

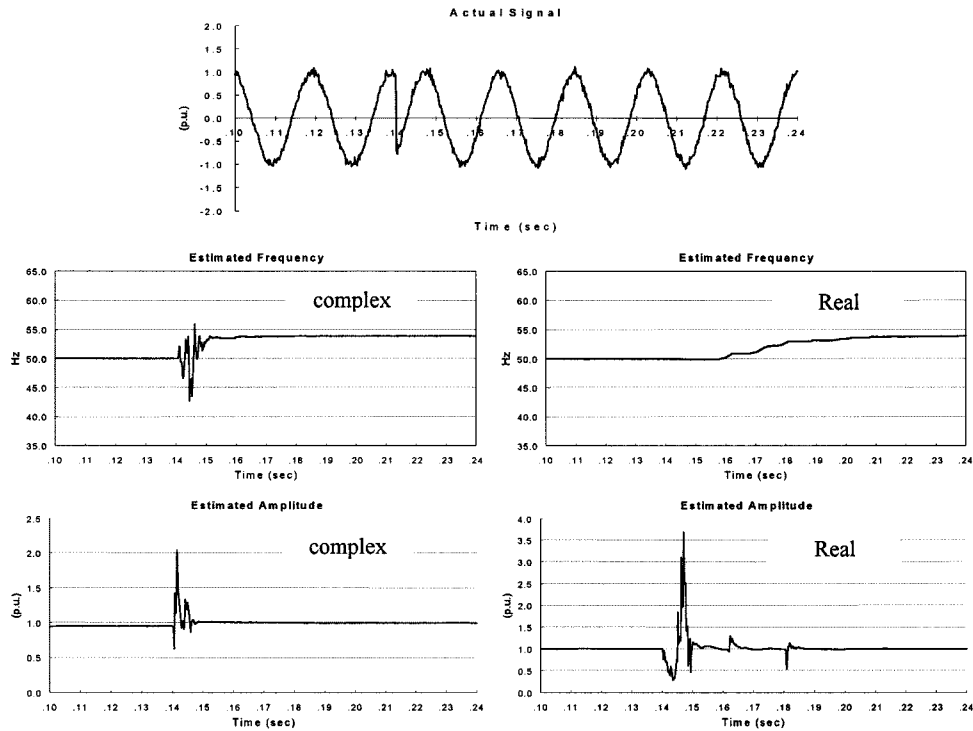


Fig. 6. An 8% step up in the frequency for signal 1.

$$\text{Signal} = A \cos(k\omega T_s + \phi_1) + \frac{A}{10} \cos(3k\omega T_s + \phi_3) + \frac{A}{20} \cos(5k\omega T_s + \phi_5) + \text{noise}$$

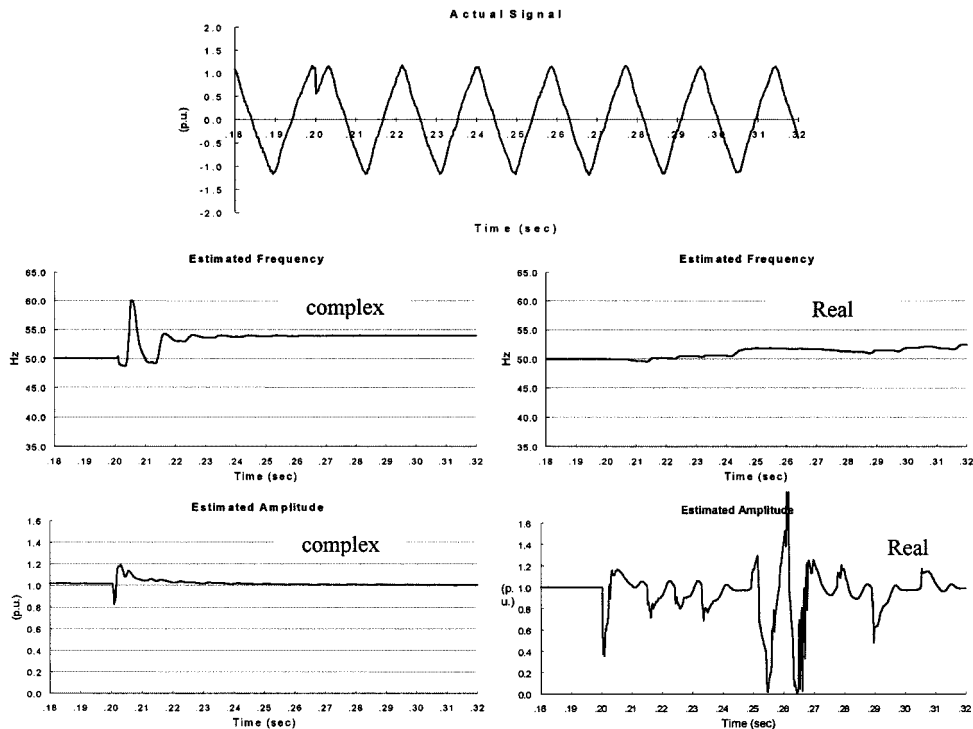


Fig. 7. An 8% step up in the frequency for signal 2.

s) of the fundamental time period. The proposed complex filter exhibits a little oscillation in the frequency during the ampli-

tude change. For the second type of signal (i.e., signal with harmonics) the performance of the complex Kalman filter is much

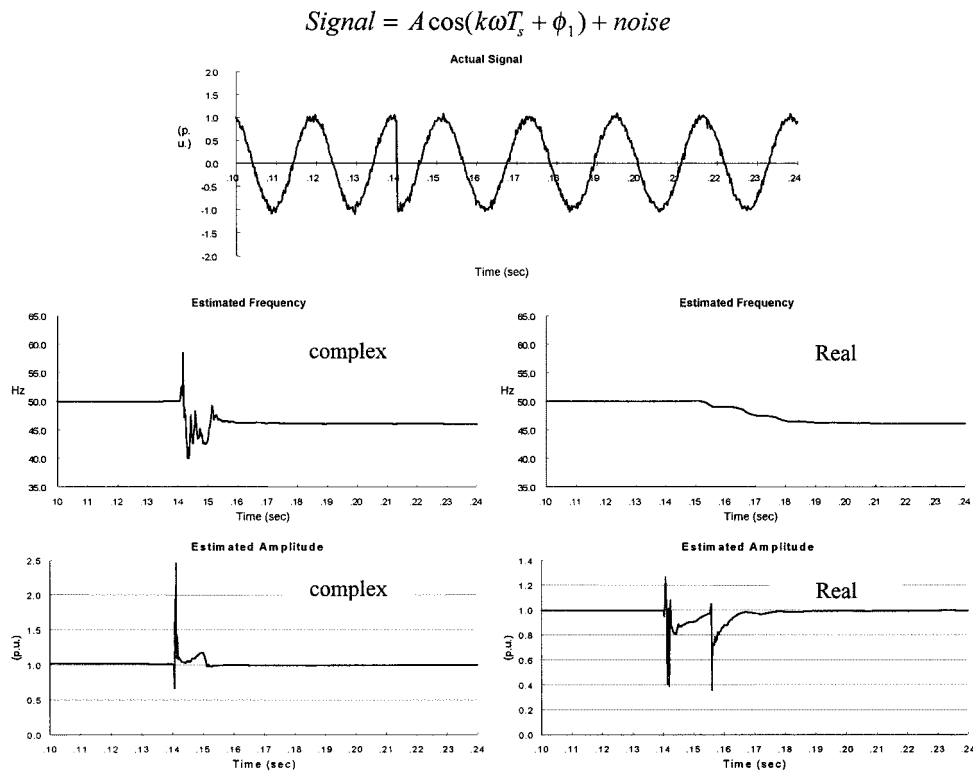


Fig. 8. Step down in frequency of signal 1.

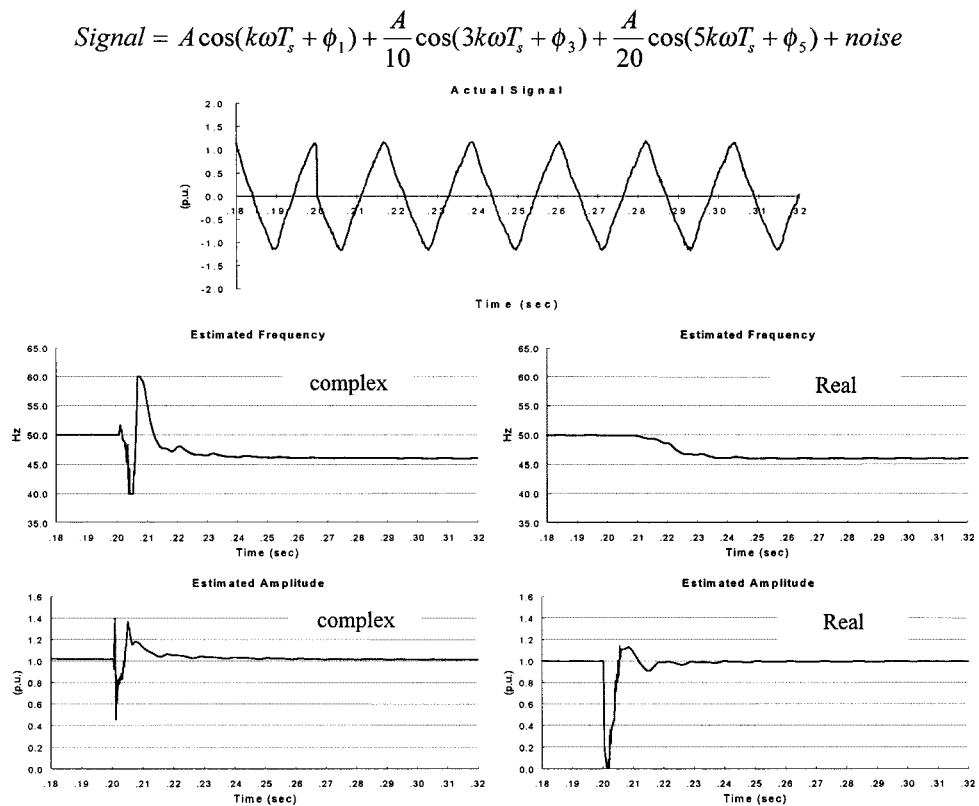


Fig. 9. Step down in frequency of signal 2.

better. In this case the tracking time is a little over half a cycle (about 0.011 s). The real Kalman filter is unable to track the amplitude with 1.2 p.u. fundamental, 0.12 p.u. third harmonic, and

0.06 p.u. fifth harmonic. However, as soon as the amplitudes of these components are reduced to 0.8, 0.08, and 0.04, respectively, the filter settles down quickly.

Case-II: In this case, the amplitude is suddenly increased from 1.0 p.u. to 1.4 p.u for both signals. The comparative study shows similar performances of the filters (Fig. 4). They almost settle within half a cycle (0.01 s) of the fundamental signal. However, a little dent in the frequency is observed in the case of the complex Kalman filter. For the test signal II the real Kalman filter loses complete track of the signal with the new value of amplitude (Fig. 5). Under identical conditions the complex Kalman filter takes almost 1-cycle (0.02 s) of the fundamental to track the amplitude, phase and frequency of the signal with the new amplitude.

Case-III: The frequency is suddenly increased from 50 Hz to 54 Hz. The study shows that the complex Kalman filter exhibits quicker tracking performance as compared to the ordinary Kalman filter in the case of the signal without harmonics (Fig. 6). The proposed filter settles within 0.75 cycle (0.015 s) of the fundamental signal. But the real Kalman filter takes over five cycles. For the test signal-II, the latter becomes unstable after the frequency has been increased, whereas the proposed filter takes around three cycles to settle down at the new value of the frequency. These waveforms are depicted in Fig. 7.

Case-IV: Fig. 8 shows the waveforms when the frequency is suddenly reduced from 50 Hz to 46 Hz. For test signal-I the proposed filter takes around 0.015 s (0.75 cycles) to settle at the new value of the frequency. The real filter is slower and takes almost one and a half cycles (0.03 s) before settling at these values. For the test signal-II the real filter malfunctions and fails to track after the frequency is changed. But the proposed complex filter settles within one and a half cycles (Fig. 9).

Thus, the above results exhibit better performance of the complex filter in estimating the signal frequency. Besides it is observed that the real filter malfunctions and loses complete track of the signal when the signal is contaminated with harmonics and noise.

IV. CONCLUSION

A nonlinear filter based on the complex Kalman filter has been proposed. The hysteresis method has been suggested for resetting the covariance matrix, which enables fast tracking of the frequency. The proposed filter offers superior performances in all cases. In addition the filter involves less computation which makes it attractive for real-time implementation. The issues such as stability of the algorithm and performance under other disturbances are currently under investigation.

APPENDIX

SIGNAL MODEL FOR REAL EXTENDED KALMAN FILTER

Let the signal model be represented by

$$z_k = a_1 \sin(kw_1T_s + \phi_1) \quad (A1)$$

where T_s = sampling time, and ω_1 = the fundamental angular frequency.

The observation signal can be represented in a nonlinear state-space model as follows:

$$\begin{bmatrix} x_{1(k+1)} \\ x_{2(k+1)} \\ x_{3(k+1)} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_{1(k)} \\ x_{2(k)} \\ x_{3(k)} \end{bmatrix} \quad (A2)$$

where

$$x_1 = a_1; \quad x_2 = w_1; \quad x_3 = \phi_1 \quad (A3)$$

$$y_k = x_{1(k)} \sin(kx_{2(k)}T_s + x_{3(k)}) + v_k \quad (A4)$$

$$v_k = \text{measurement noise}$$

The above nonlinear process can be represented as

$$x_{k+1} = x_k \quad (A5)$$

$$y_{k+1} = G(x_k) + v_k \quad (A6)$$

Linearizing the above system and applying the extended Kalman filter to the first-order system, a nonlinear recursive filter for estimating a single sinusoid and its frequency in white noise is obtained as follows [7]:

$$\hat{x}_{k|k} = \hat{x}_{k|k-1} + K_k(y_k - G(\hat{x}_{k|k-1})) \quad (A7)$$

$$\hat{x}_{k+1|k} = \hat{x}_{k|k} \quad (A8)$$

$$K_k = \hat{P}_{k|k-1} \left. \frac{\partial G^T}{\partial X} \right|_{(k,k-1)} \left[\left. \frac{\partial G}{\partial X} \right|_{(k,k-1)} \hat{P}_{k|k-1} \cdot \left. \frac{\partial G^T}{\partial X} \right|_{(k,k-1)} + 1 \right]^{-1} \quad (A9)$$

$$\hat{P}_{k|k} = \hat{P}_{k|k-1} - K_k \left. \frac{\partial G}{\partial X} \right|_{(k,k-1)} \hat{P}_{k|k-1} \quad (A10)$$

$$\hat{P}_{k|k+1} = \hat{P}_{k|k} \quad (A11)$$

The initial estimate of the covariance matrix P is decided by trial and error and fixed to an identity matrix. The covariance resetting under abnormal changing conditions is decided by the same hysteresis method discussed earlier. The filter is nonlinear and therefore the gain K_k and the covariance matrix $\hat{P}_{k|k}$ depend on the estimate $\hat{x}_{k|k}$ of the state vector x_k .

REFERENCES

- [1] D. Hart *et al.*, "A new frequency tracking and phasor estimation algorithm for generator protection," *IEEE Trans. Power Delivery*, vol. 12, July 1997.
- [2] J. F. Chicharo *et al.*, "Power system harmonic signal estimation and retrieval for active power filter applications," *IEEE Trans. Power Electron.*, vol. 9, pp. 580–585, Nov. 1994.
- [3] Moore *et al.*, "Model system tests on a new numeric method of power system frequency measurement," *IEEE Trans. Power Delivery*, vol. 11, Apr.(PAGE NOS?) 1996.
- [4] B. F. La Scala *et al.*, "Design of an extended Kalman filter frequency tracker," *IEEE Trans. Signal Processing*, vol. 44, pp. 739–742, Mar. 1996.
- [5] P. Tichavsky *et al.*, "Comparative study of four adaptive frequency trackers," *IEEE Trans. Signal Processing*, vol. 45, pp. 1473–1484, June 1997.
- [6] K. Nishiyama, "A nonlinear filter for estimating a sinusoidal signal and its parameters in white noise: On the case of a single sinusoid," *IEEE Trans. Signal Processing*, vol. 45, pp. 970–981, Apr. 1997.
- [7] C. K. Chui and G. Chen, *Kalman Filtering with Real-Time Applications*, 2nd ed. New York: Springer-Verlag.

P. K. Dash was educated at the Utkal University and the Indian Institute of Science, Bangalore.

He was a Post-Doctoral Fellow at the University of Calgary, Calgary, Alta., Canada, and had several visiting appointments with North American universities, BBC Brown Boveri, Switzerland, and Bristol Aerospace, Canada. His recent collaborations are with Virginia Polytechnic Institute and State University, Blacksburg. He is a Professor of Electrical Engineering and Chairman of the Centre for Applied Artificial Intelligence, Regional Engineering College, Rourkela, India. During 1993–1994 and 1996–1997, he was a Visiting Staff Member at National University of Singapore.

G. Panda received the first class degree (Honors) in electrical engineering in 1971 and the M.Sc.Engg. degree in communication systems in 1977 from Sambalpur University, India, and the Ph.D. degree in “digital signal processing” in 1982 from Indian Institute of Technology, Kharagpur. He was awarded Common Wealth Academic Fellowship and pursued postdoctoral work in adaptive signal processing at the University of Edinburgh, Edinburgh, U.K., during 1984–1986.

He was a Lecturer during 1971–1979, and a Reader during 1979–1988 at University College of Engineering, Burla, India. In 1988, he joined Regional Engineering, Rourkela, India, as Professor and acted as Founder Head in Department of Applied Electronics and Instrumentation Engineering. His research area includes data communication, channel equalization, and ANN application to communication systems.

R. K. Jena is currently pursuing the Ph.D. degree in the Department of Electrical Engineering at the Regional Engineering College, Rourkela, India. His research interests include power quality measurements and intelligent system applications.

Aurobinda Routray is an Assistant Professor of electrical engineering at the Indian Institute of Technology, Kharagpur. His research interests include industrial electronics, automatic control and electric vehicles, and power electronics, including HVDC transmission.